Two-tail Z test ↔ confidence interval

A1S2: SLIDES 1-51 ONLY

1 RECOGNISE IT

- Two-tail mean test at α and matching confidence coefficient 1-α.
- Population σ known.
- Asked to compare test and CI conclusions.

2 DECISION / METHOD

1. Build the matching Z CI: $\bar{X} \pm z^*SE$.
2. If μ_0 lies outside CI -> reject H_0:$\mu=\mu_0$.
3. If μ_0 lies inside CI -> do not reject H_0.
4. This equivalence is for matching two-tail α/confidence.

3 FORMULA

$z^* = \Phi^{-1}(1-\alpha/2) > 0$

$CI: \bar{X} \pm z^* \times \sigma/\sqrt{n}$

two-tail reject H_0 iff μ_0 is outside CI

4 FORMULA SHEET VS WHAT I KNOW

Gives: Z SE lane.
Still know: match confidence coefficient 1-α and distinguish a mean CI from an individual-value range.

5 EXCEL ROUTE

- z=NORM.S.INV(1-alpha/2)
- lower/upper=$\bar{X}$ ± z*σ/SQRT(n)
- =IF(OR(mu0<lower,mu0>upper),"Reject H0","Do not reject H0").

6 TINY WORKED EXAMPLE

$\bar{X}$=52,σ=8,n=64,α=.05 gives CI (50.04,53.96). μ_0=50 is outside -> reject H_0 in the matching two-tail test.

7 TOP EXAM TRAPS

- Using a 90% CI for α=.05.
- Applying equivalence to a one-tail test.
- Saying values outside CI are impossible observations.

8 SAY ALOUD

For a matching two-tail test, reject exactly when the hypothesised mean is outside the confidence interval.

9 EXIT CHECK

At α=.01, which confidence level must the comparison interval use?

ONE-GLANCE VISUAL ROUTE

